

Principles of Programming Languages Paradigms: Object Oriented Programming.

Andrei Arusoaie¹

¹Department of Computer Science

December 14, 2021

Outline

Paradigms

Outline

Paradigms

Object-Oriented Programming

Outline

Paradigms

Object-Oriented Programming

Conclusion

Paradigms

- ▶ Imperative Programming: ✓

Paradigms

- ▶ Imperative Programming: ✓
- ▶ Object Oriented Programming

Paradigms

- ▶ Imperative Programming: ✓
- ▶ Object Oriented Programming
- ▶ Declarative programming
 - ▶ Functional Programming

Paradigms

- ▶ Imperative Programming: ✓
- ▶ Object Oriented Programming
- ▶ Declarative programming
 - ▶ Functional Programming
 - ▶ Logic Programming

Object-Oriented Programming: Motivation-1

- ▶ **Complex** software

Object-Oriented Programming: Motivation-1

- ▶ **Complex** software
- ▶ NATO conference, 1968: *software crisis*
 1. User dissatisfaction with the final product

Object-Oriented Programming: Motivation-1

- ▶ **Complex** software
- ▶ NATO conference, 1968: *software crisis*
 1. User dissatisfaction with the final product
 2. cost overruns

Object-Oriented Programming: Motivation-1

- ▶ **Complex** software
- ▶ NATO conference, 1968: *software crisis*
 1. User dissatisfaction with the final product
 2. cost overruns
 3. buggy software

Object-Oriented Programming: Motivation-1

- ▶ **Complex** software
- ▶ NATO conference, 1968: *software crisis*
 1. User dissatisfaction with the final product
 2. cost overruns
 3. buggy software
 4. brittle software

Motivation-2

Main issues in large software projects:

- ▶ Names: functions, variables, ...

Motivation-2

Main issues in large software projects:

- ▶ Names: functions, variables, ...
- ▶ Constraints: time and space complexity

Motivation-2

Main issues in large software projects:

- ▶ Names: functions, variables, ...
- ▶ Constraints: time and space complexity
- ▶ Memory management (garbage collection and address spaces)

Motivation-2

Main issues in large software projects:

- ▶ Names: functions, variables, ...
- ▶ Constraints: time and space complexity
- ▶ Memory management (garbage collection and address spaces)
- ▶ Concurrency

Motivation-2

Main issues in large software projects:

- ▶ Names: functions, variables, ...
- ▶ Constraints: time and space complexity
- ▶ Memory management (garbage collection and address spaces)
- ▶ Concurrency
- ▶ Event-driven interfaces

Motivation-2

Main issues in large software projects:

- ▶ Names: functions, variables, ...
- ▶ Constraints: time and space complexity
- ▶ Memory management (garbage collection and address spaces)
- ▶ Concurrency
- ▶ Event-driven interfaces

How to fix all these?

How to fix all these?

- ▶ Extreme discipline

How to fix all these?

- ▶ Extreme discipline (hard to get when working in teams)

How to fix all these?

- ▶ Extreme discipline (hard to get when working in teams)
- ▶ Or a new paradigm that facilitates:

How to fix all these?

- ▶ Extreme discipline (hard to get when working in teams)
- ▶ Or a new paradigm that facilitates:
 1. solving very complex problems

How to fix all these?

- ▶ Extreme discipline (hard to get when working in teams)
- ▶ Or a new paradigm that facilitates:
 1. solving very complex problems
 2. working in teams

How to fix all these?

- ▶ Extreme discipline (hard to get when working in teams)
- ▶ Or a new paradigm that facilitates:
 1. solving very complex problems
 2. working in teams
 3. code modularity

How to fix all these?

- ▶ Extreme discipline (hard to get when working in teams)
- ▶ Or a new paradigm that facilitates:
 1. solving very complex problems
 2. working in teams
 3. code modularity
 4. easy development and code maintenance

Object-Oriented Programming

A bunch of concepts:

Object-Oriented Programming

A bunch of concepts:

- ▶ Classes, objects, members (attributes + methods), access modifiers

Object-Oriented Programming

A bunch of concepts:

- ▶ Classes, objects, members (attributes + methods), access modifiers
- ▶ Constructors, destructors

Object-Oriented Programming

A bunch of concepts:

- ▶ Classes, objects, members (attributes + methods), access modifiers
- ▶ Constructors, destructors
- ▶ Overloading, overriding, shadowing

Object-Oriented Programming

A bunch of concepts:

- ▶ Classes, objects, members (attributes + methods), access modifiers
- ▶ Constructors, destructors
- ▶ Overloading, overriding, shadowing
- ▶ *Encapsulation*, information hiding

Object-Oriented Programming

A bunch of concepts:

- ▶ Classes, objects, members (attributes + methods), access modifiers
- ▶ Constructors, destructors
- ▶ Overloading, overriding, shadowing
- ▶ *Encapsulation*, information hiding
- ▶ *Abstraction*

Object-Oriented Programming

A bunch of concepts:

- ▶ Classes, objects, members (attributes + methods), access modifiers
- ▶ Constructors, destructors
- ▶ Overloading, overriding, shadowing
- ▶ *Encapsulation*, information hiding
- ▶ *Abstraction*
- ▶ *Inheritance*

Object-Oriented Programming

A bunch of concepts:

- ▶ Classes, objects, members (attributes + methods), access modifiers
- ▶ Constructors, destructors
- ▶ Overloading, overriding, shadowing
- ▶ *Encapsulation*, information hiding
- ▶ *Abstraction*
- ▶ *Inheritance*
- ▶ *Polymorphism*

Object-Oriented Programming

A bunch of concepts:

- ▶ Classes, objects, members (attributes + methods), access modifiers
- ▶ Constructors, destructors
- ▶ Overloading, overriding, shadowing
- ▶ *Encapsulation*, information hiding
- ▶ *Abstraction*
- ▶ *Inheritance*
- ▶ *Polymorphism*
- ▶ Dynamic method selection (or dynamic dispatch)

Object-Oriented Programming

A bunch of concepts:

- ▶ Classes, objects, members (attributes + methods), access modifiers
- ▶ Constructors, destructors
- ▶ Overloading, overriding, shadowing
- ▶ *Encapsulation*, information hiding
- ▶ *Abstraction*
- ▶ *Inheritance*
- ▶ *Polymorphism*
- ▶ Dynamic method selection (or dynamic dispatch)
- ▶ Interfaces, abstract classes

Classes vs. Objects

► Class =

Classes vs. Objects

- ▶ Class = a **specification** used to *construct* objects

Classes vs. Objects

- ▶ Class = a **specification** used to *construct* objects
- ▶ Classes establish the name of the class, the *data method* members with types and signature, visibility and implementation.

Classes vs. Objects

- ▶ Class = a **specification** used to *construct* objects
- ▶ Classes establish the name of the class, the *data method* members with types and signature, visibility and implementation.
- ▶ Object =

Classes vs. Objects

- ▶ Class = a **specification** used to *construct* objects
- ▶ Classes establish the name of the class, the *data method* members with types and signature, visibility and implementation.
- ▶ Object = an *instance* of a class; it has a state where its *data members* (or *attributes* or *properties*) – described in the corresponding class – have values

Constructors and Destructors

- ▶ *Constructor*

Constructors and Destructors

- ▶ *Constructor*: a special “method” which has the same name as the class and no return type

Constructors and Destructors

- ▶ *Constructor*: a special “method” which has the same name as the class and no return type
- ▶ When an object is created the *constructor* is called: allocate the necessary memory, initialise members (do not forget inheritance)

Constructors and Destructors

- ▶ *Constructor*: a special “method” which has the same name as the class and no return type
- ▶ When an object is created the *constructor* is called: allocate the necessary memory, initialise members (do not forget inheritance)
- ▶ Typically (in many OO programming languages) we use the `new` keyword to call the constructor

Constructors and Destructors

- ▶ *Constructor*: a special “method” which has the same name as the class and no return type
- ▶ When an object is created the *constructor* is called: allocate the necessary memory, initialise members (do not forget inheritance)
- ▶ Typically (in many OO programming languages) we use the `new` keyword to call the constructor
- ▶ *Destructor*

Constructors and Destructors

- ▶ *Constructor*: a special “method” which has the same name as the class and no return type
- ▶ When an object is created the *constructor* is called: allocate the necessary memory, initialise members (do not forget inheritance)
- ▶ Typically (in many OO programming languages) we use the `new` keyword to call the constructor
- ▶ *Destructor*: frees the memory occupied by the object

Constructors and Destructors

- ▶ *Constructor*: a special “method” which has the same name as the class and no return type
- ▶ When an object is created the *constructor* is called: allocate the necessary memory, initialise members (do not forget inheritance)
- ▶ Typically (in many OO programming languages) we use the `new` keyword to call the constructor
- ▶ *Destructor*: frees the memory occupied by the object
- ▶ *Garbage Collector*

Constructors and Destructors

- ▶ *Constructor*: a special “method” which has the same name as the class and no return type
- ▶ When an object is created the *constructor* is called: allocate the necessary memory, initialise members (do not forget inheritance)
- ▶ Typically (in many OO programming languages) we use the `new` keyword to call the constructor
- ▶ *Destructor*: frees the memory occupied by the object
- ▶ *Garbage Collector*: automatic memory management (efficient memory allocation, automatically frees the memory, enables memory reuse)

Encapsulation

► *Encapsulation*

Encapsulation

- ▶ *Encapsulation*: organise the data and operations (over that data) in a single unit

Encapsulation

- ▶ *Encapsulation*: organise the data and operations (over that data) in a single unit
- ▶ Advantage: manipulate the objects in an uniform and consistent way using methods

Encapsulation

- ▶ *Encapsulation*: organise the data and operations (over that data) in a single unit
- ▶ Advantage: manipulate the objects in an uniform and consistent way using methods
- ▶ *Information hiding*: separate the interface (i.e., the public information in a class) from the implementation

Encapsulation

- ▶ *Encapsulation*: organise the data and operations (over that data) in a single unit
- ▶ Advantage: manipulate the objects in an uniform and consistent way using methods
- ▶ *Information hiding*: separate the interface (i.e., the public information in a class) from the implementation
- ▶ *Access modifiers*: used to restrict/grant access to object's data

Abstraction

- ▶ It comes in the same “package” with *Encapsulation*

Abstraction

- ▶ It comes in the same “package” with *Encapsulation*
- ▶ Example: driving a car doesn't require knowing the internals of the combustion engine

Abstraction

- ▶ It comes in the same “package” with *Encapsulation*
- ▶ Example: driving a car doesn't require knowing the internals of the combustion engine
- ▶ *Abstraction*: preserve the information that is important in a given context

Abstraction

- ▶ It comes in the same “package” with *Encapsulation*
- ▶ Example: driving a car doesn't require knowing the internals of the combustion engine
- ▶ *Abstraction*: preserve the information that is important in a given context
- ▶ The human mind can operate easier with abstractions

Abstraction

- ▶ It comes in the same “package” with *Encapsulation*
- ▶ Example: driving a car doesn't require knowing the internals of the combustion engine
- ▶ *Abstraction*: preserve the information that is important in a given context
- ▶ The human mind can operate easier with abstractions
- ▶ **Experiment:**

Abstraction

- ▶ It comes in the same “package” with *Encapsulation*
- ▶ Example: driving a car doesn't require knowing the internals of the combustion engine
- ▶ *Abstraction*: preserve the information that is important in a given context
- ▶ The human mind can operate easier with abstractions
- ▶ **Experiment**: 197328643791468251973286437914682 (30 sec)

Abstraction

- ▶ It comes in the same “package” with *Encapsulation*
- ▶ Example: driving a car doesn't require knowing the internals of the combustion engine
- ▶ *Abstraction*: preserve the information that is important in a given context
- ▶ The human mind can operate easier with abstractions
- ▶ **Experiment**: 197328643791468251973286437914682 (30 sec)
 1. How many can you remember?
 2. How many can you remember with unlimited amount of time?

Inheritance

► *Inheritance*

Inheritance

- ▶ *Inheritance*: reuse the code/features from a previously defined class

Inheritance

- ▶ *Inheritance*: reuse the code/features from a previously defined class
- ▶ Principle: DRY – “Do not repeat yourself”

Inheritance

- ▶ *Inheritance*: reuse the code/features from a previously defined class
- ▶ Principle: DRY – “Do not repeat yourself”
- ▶ If class `Child` inherits from class `Parent` then `Child` uses the same implementation as `Parent`...

Inheritance

- ▶ *Inheritance*: reuse the code/features from a previously defined class
- ▶ Principle: DRY – “Do not repeat yourself”
- ▶ If class `Child` inherits from class `Parent` then `Child` uses the same implementation as `Parent`...
- ▶ ... with some exceptions:

Inheritance

- ▶ *Inheritance*: reuse the code/features from a previously defined class
- ▶ Principle: DRY – “Do not repeat yourself”
- ▶ If class `Child` inherits from class `Parent` then `Child` uses the same implementation as `Parent`...
- ▶ ... with some exceptions:
 - ▶ *Field hiding*: `Child` can redefine data members (i.e, use the same name) from `Parent`

Inheritance

- ▶ *Inheritance*: reuse the code/features from a previously defined class
- ▶ Principle: DRY – “Do not repeat yourself”
- ▶ If class `Child` inherits from class `Parent` then `Child` uses the same implementation as `Parent`...
- ▶ ... with some exceptions:
 - ▶ *Field hiding*: `Child` can redefine data members (i.e, use the same name) from `Parent`
 - ▶ *Overriding*: `Child` can override methods defined in `Parent`

Inheritance

- ▶ *Inheritance*: reuse the code/features from a previously defined class
- ▶ Principle: DRY – “Do not repeat yourself”
- ▶ If class `Child` inherits from class `Parent` then `Child` uses the same implementation as `Parent`...
- ▶ ... with some exceptions:
 - ▶ *Field hiding*: `Child` can redefine data members (i.e, use the same name) from `Parent`
 - ▶ *Overriding*: `Child` can override methods defined in `Parent`
- ▶ Single vs. multiple inheritance

Polymorphism

Polymorphism

- ▶ It comes in many flavours:

Polymorphism

- ▶ It comes in many flavours:
 1. Overloading: multiple implementations of the same method with different parameters

Polymorphism

- ▶ It comes in many flavours:
 1. Overloading: multiple implementations of the same method with different parameters
 2. Parametric – often used with *generics*: methods work with parametric classes (e.g., `sort` for `List<T>`)

Polymorphism

- ▶ It comes in many flavours:
 1. Overloading: multiple implementations of the same method with different parameters
 2. Parametric – often used with *generics*: methods work with parametric classes (e.g., `sort` for `List<T>`)
 3. Subtype: single interface to multiple representations of the same type (the most common type of polymorphism)

Polymorphism

- ▶ It comes in many flavours:
 1. Overloading: multiple implementations of the same method with different parameters
 2. Parametric – often used with *generics*: methods work with parametric classes (e.g., `sort` for `List<T>`)
 3. Subtype: single interface to multiple representations of the same type (the most common type of polymorphism)
- ▶ Subtype polymorphism: uses inheritance

Subtype polymorphism - Liskov Substitution Principle

- ▶ Suppose that `Child` inherits `Parent`, and consider a function `SomeType foo(Parent p) {...}`

Subtype polymorphism - Liskov Substitution Principle

- ▶ Suppose that `Child` inherits `Parent`, and consider a function `SomeType foo(Parent p) {...}`
- ▶ In this case `foo` can be applied to an instance of *any* subclass of `Parent`:

```
Child child = new Child();  
SomeType t = foo(child);
```

Subtype polymorphism - Liskov Substitution Principle

- ▶ Suppose that `Child` inherits `Parent`, and consider a function `SomeType foo(Parent p) {...}`
- ▶ In this case `foo` can be applied to an instance of *any* subclass of `Parent`:

```
Child child = new Child();  
SomeType t = foo(child);
```

- ▶ Consider `Parent bar(Parent p) { return p; }`:
`Child child = new Child();`
`Child p = bar(child);`

Subtype polymorphism - Liskov Substitution Principle

- ▶ Suppose that `Child` inherits `Parent`, and consider a function `SomeType foo(Parent p) {...}`
- ▶ In this case `foo` can be applied to an instance of *any* subclass of `Parent`:

```
Child child = new Child();  
SomeType t = foo(child);
```

- ▶ Consider `Parent bar(Parent p) { return p; }`:
`Child child = new Child();`
`Child p = bar(child); // rejected by static type checker`

Subtype polymorphism - Liskov Substitution Principle

- ▶ Suppose that `Child` inherits `Parent`, and consider a function `SomeType foo(Parent p) {...}`
- ▶ In this case `foo` can be applied to an instance of *any* subclass of `Parent`:

```
Child child = new Child();  
SomeType t = foo(child);
```

- ▶ Consider `Parent bar(Parent p) { return p; }`:
`Child child = new Child();`
`Child p = bar(child); // rejected by static type checker`
`Child p = (Child) bar(child);`

Subtype polymorphism - Liskov Substitution Principle

- ▶ Suppose that `Child` inherits `Parent`, and consider a function `SomeType foo(Parent p) {...}`
- ▶ In this case `foo` can be applied to an instance of *any* subclass of `Parent`:

```
Child child = new Child();  
SomeType t = foo(child);
```

- ▶ Consider `Parent bar(Parent p) { return p; }`:
`Child child = new Child();`
`Child p = bar(child); // rejected by static type checker`
`Child p = (Child) bar(child); // dynamic check`

Dynamic dispatch

- ▶ *Dynamic dispatch*

Dynamic dispatch

- ▶ *Dynamic dispatch*: select the correct implementation of a polymorphic method at runtime

Dynamic dispatch

- *Dynamic dispatch*: select the correct implementation of a polymorphic method at runtime

```
class Parent {... void f() {...} ...}  
class Child : Parent {... void f() {...} ...}  
...  
Child child = new Child();  
child.f()
```

Dynamic dispatch

- *Dynamic dispatch*: select the correct implementation of a polymorphic method at runtime

```
class Parent {... void f() {...} ...}  
class Child : Parent {... void f() {...} ...}  
...  
Child child = new Child();  
child.f() // f
```

Dynamic dispatch

- *Dynamic dispatch*: select the correct implementation of a polymorphic method at runtime

```
class Parent {... void f() {...} ...}  
class Child : Parent {... void f() {...} ...}  
...  
Child child = new Child();  
child.f() // f  
(Parent) child.f();
```

Dynamic dispatch

- *Dynamic dispatch*: select the correct implementation of a polymorphic method at runtime

```
class Parent {... void f() {...} ...}  
class Child : Parent {... void f() {...} ...}  
...  
Child child = new Child();  
child.f() // f  
(Parent) child.f(); // f
```

Late binding

- ▶ *Late binding*

Late binding

► *Late binding:*

```
class Parent {  
    int a = 0;  
    void f() { g(); }  
    void g() { a = 1; }  
}  
class Child : Parent {  
    int b = 0;  
    void g() { b = 2; }  
}  
...  
Child child = new Child();  
child.f()
```

Late binding

► *Late binding:*

```
class Parent {  
    int a = 0;  
    void f() { g(); }  
    void g() { a = 1; }  
}  
class Child : Parent {  
    int b = 0;  
    void g() { b = 2; }  
}  
...  
Child child = new Child();  
child.f() // b =
```

Late binding

► *Late binding:*

```
class Parent {  
    int a = 0;  
    void f() { g(); }  
    void g() { a = 1; }  
}  
class Child : Parent {  
    int b = 0;  
    void g() { b = 2; }  
}  
...  
Child child = new Child();  
child.f() // b = 2;
```

Late binding

► *Late binding:*

```
class Parent {  
    int a = 0;  
    void f() { g(); }  
    void g() { a = 1; }  
}  
class Child : Parent {  
    int b = 0;  
    void g() { b = 2; }  
}  
...  
Child child = new Child();  
child.f() // b = 2; a =
```

Late binding

► *Late binding:*

```
class Parent {  
    int a = 0;  
    void f() { g(); }  
    void g() { a = 1; }  
}  
class Child : Parent {  
    int b = 0;  
    void g() { b = 2; }  
}  
...  
Child child = new Child();  
child.f() // b = 2; a = 0;
```

Interfaces vs. Abstract classes

- ▶ *Abstract class*

Interfaces vs. Abstract classes

- ▶ *Abstract class*: class without implementation for (at least one) methods

Interfaces vs. Abstract classes

- ▶ *Abstract class*: class without implementation for (at least one) methods
- ▶ *Interface*

Interfaces vs. Abstract classes

- ▶ *Abstract class*: class without implementation for (at least one) methods
- ▶ *Interface*: has methods without implementation
- ▶ Traits:
 1. Both can be inherited

Interfaces vs. Abstract classes

- ▶ *Abstract class*: class without implementation for (at least one) methods
- ▶ *Interface*: has methods without implementation
- ▶ Traits:
 1. Both can be inherited
 2. Abstract classes can contain implementation for functions

Interfaces vs. Abstract classes

- ▶ *Abstract class*: class without implementation for (at least one) methods
- ▶ *Interface*: has methods without implementation
- ▶ Traits:
 1. Both can be inherited
 2. Abstract classes can contain implementation for functions
 3. Methods in interfaces must be implemented by the implementing class

Recap

Concepts:

- ▶ classes, attributes/properties, methods
- ▶ access modifiers
- ▶ objects
- ▶ constructors, destructors
- ▶ encapsulation
- ▶ abstraction
- ▶ inheritance
- ▶ polymorphism (overloading, parametric, subtype)
- ▶ dynamic dispatch/late binding
- ▶ interfaces, abstract classes

Claimed advantages

- ▶ Re-usability
- ▶ Redundancy
- ▶ Easy maintenance
- ▶ Modularity
- ▶ Productivity
- ▶ Lower costs
- ▶ Higher-quality software
- ▶ Security (data hiding)

Disadvantages: what people complain about

- ▶ Size of the programs
- ▶ Effort
- ▶ Speed
- ▶ Not suitable for all types of problems
- ▶ Steep learning curve

Facts

- ▶ OOP scales; it is widely used in the industry
- ▶ Not all languages that are advertised as OOP languages support all the features above (e.g., C++ does not have GC)
- ▶ Sometimes, misunderstanding of the OOP concepts may complicate your code

Resources

- ▶ Chapter 10 in **Programming Languages: Principles and Paradigms** by Maurizio Gabbrielli and Simone Martini.

URL: `http://websrv.dthu.edu.vn/attachments/newsevents/content2415/Programming\_Languages\_-\_Principles\_and\_Paradigms\_thereads1106.pdf`